

Laser Cutting

Week: 4 Laser-Cutting.

Introduction

This week, I was introduced to **laser cutting**, a digital fabrication process used to cut and engrave materials using a focused laser beam. Before operating the machine, I learned about the basic working principles of a CO₂ laser cutter, the importance of machine settings, and the safety precautions that need to be followed during operation.

The laser cutter available in our lab is a **1490 CO₂ Laser Cutter** with a working area of **1y300 × 900 mm** and a **150 W laser power**. The machine is capable of cutting and engraving materials such as acrylic, plywood, MDF, cardboard and paper.



Specifications:

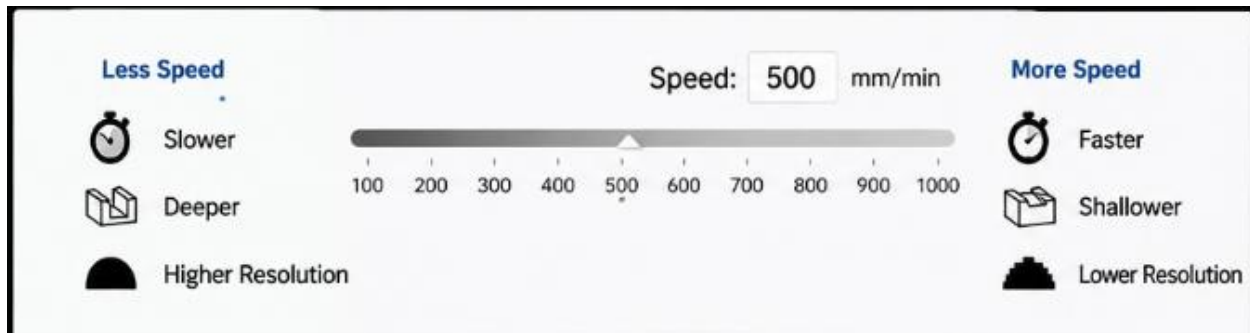
LASER CUTTER



- **Model:** 1490 Co2 laser
- **Working Area:** 1300 x 900mm
- **Laser power:** 150W
- **Cutting Speed:** 25m/min
- **Engraving Speed:** 55m/min
- **Accuracy:** 0.1mm
- **Machine Power:** 1000W
- **Working Temperature:** 0°C-40°C
- **Blowing System:** Lower Blowing System
- **Applicable Material:** Acrylic, Plywood, MDF, Foam Board, Cardboard, Paper

Basics Of Laser Cutting Machine:

In laser cutting, understanding the relationship between laser speed and cut quality is essential. When the laser moves at a slower speed, it cuts deeper and produces higher resolution. In contrast, at a faster speed, the cut becomes shallower with lower resolution, affecting the precision and depth of the output.



Core Concepts:

Laser Cutting Test – Effect of Speed, Power and Quality										
Speed (mm/min) Power: 150W	Speed: 100	Speed: 200	Speed: 300	Speed: 400	Speed: 500	Speed: 600	Speed: 700	Speed: 800	Speed: 900	Speed: 1000
Cut Line (Plywood 3mm)										
Engraving (Power: 150W)										
Cut Through (Acrylic 3mm)										

Note: Results may vary based on material type, thickness, focus position and environmental conditions.

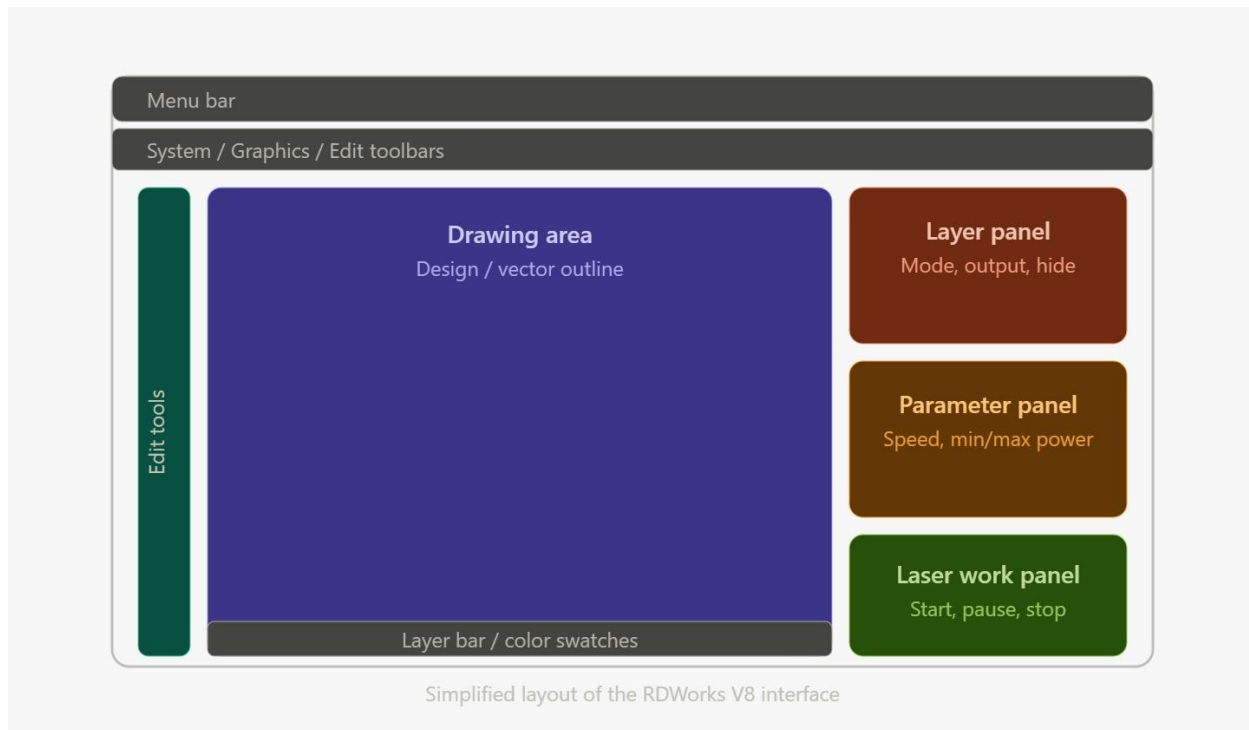
What I Learned Before Cutting

Before starting the actual cutting process, I learned that the quality of a laser-cut design depends mainly on parameters such as **laser power and speed**. The correct combination of these settings is important for obtaining a clean and accurate cut.

I also learned about the basic safety requirements while working with a CO₂ laser cutter. Proper ventilation, cooling, air assistance and electrical safety are important because the machine uses a high-power laser and produces fumes during cutting.

Another important part of the process was understanding how a digital design is converted into instructions that the laser cutter can understand.

RDWorks V8



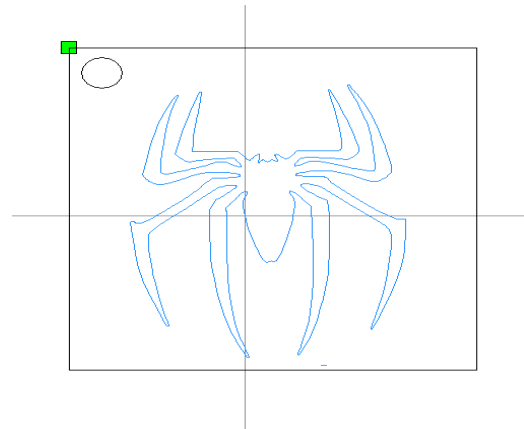
For this process, I used **RDWorks V8**, the software used to control the laser cutting machine.

I learned how to:

- Import and work with vector designs
- Position the design within the machine's working area.
- Check the cutting paths.
- Set different laser parameters.
- Prepare the design for the cutting operation.
- Send the prepared design to the laser cutter.

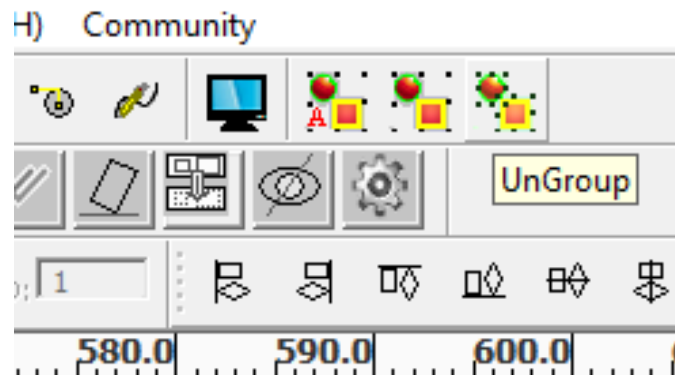
The design I worked with was a spider-shaped pattern containing multiple outlines and internal details. I prepared the design in RDWorks before sending it to the machine.

What I Learned



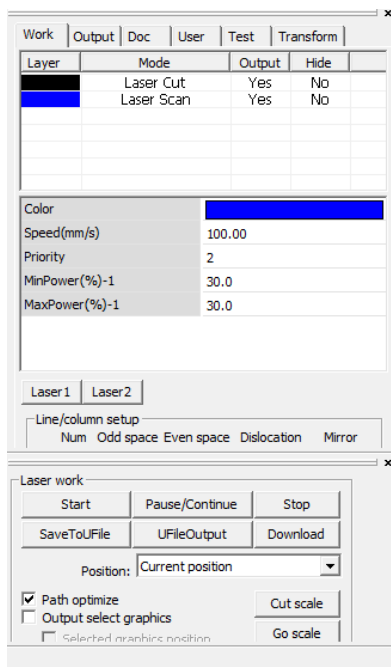
3. Separated the Image into Individual Parts

I then **unwrapped/separated the imported geometry** into individual sections. This made it easier to work with the different parts of the design separately.



4. Created a Box and Set It for 3-Cycle Cutting

Next, I created a box around the required design area and selected it for **3-cycle cutting**. This helped prepare the outer section of the material for the cutting process

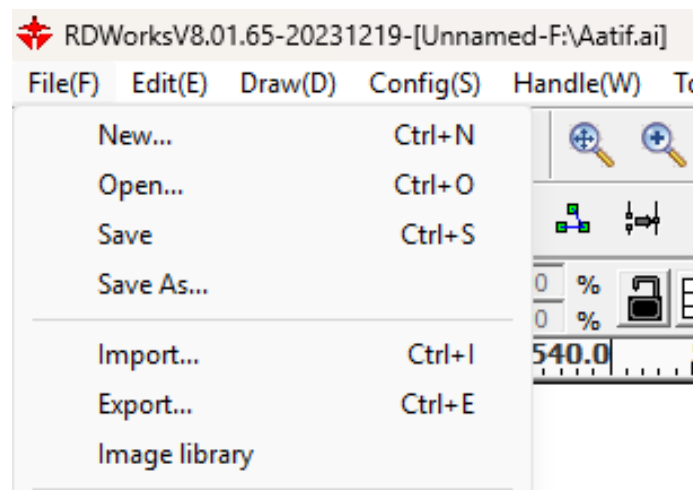


5. Set the Outline and Internal Design

After that, I selected the **outer outline** of the design and set it for a **1-cycle cut**. I also selected the internal parts of the design and set them to **Scan**, so that the internal details would be engraved rather than completely cut through.

6. Final Preparation and Export

Once the cutting and scanning settings were assigned to the different parts, I checked the complete design and made sure the paths and settings were correct. Finally, I **exported the prepared file** for the laser-cutting machine.



This experience gave me a better understanding of digital fabrication and how software and hardware work together to produce a physical design.

Laser Cutting Process

After preparing the design, I positioned it properly in the RDWorks workspace and checked the cutting paths. I then prepared the laser cutter for operation and started the cutting process. The machine followed the paths defined in RDWorks and produced the design on the material.

I observed how the laser moved along the outline and internal sections of the design to create the required shape.

The final result was a physical laser-cut version of my digital design.



My Final Output

The final output was a detailed spider-shaped laser-cut design with an outer border, internal patterns and supporting sections. The result helped me understand how a digital vector design

can be transformed into a physical object using computer-controlled fabrication.



Conclusion

This laser-cutting activity was my first practical experience with a CO₂ laser cutter. Working with the **1490 CO₂ laser cutter** and **RDWorks V8** helped me understand the complete workflow from preparing a digital design to producing the physical output.

The activity improved my understanding of **computer-controlled manufacturing, laser cutting parameters, machine operation and digital fabrication**. It also gave me practical experience in converting a digital design into a real-world object.